

CHAPTER 3 Probability

Experiment

⇒ An experiment is a process when performed will give one result only

Outcomes

⇒ The possible results of an experiment

Sample space

⇒ The set of all possible outcomes of an experiment

⇒ Is denoted by **S**

$$\{ \} \quad S = \{ H, T \}$$

Sample point

⇒ The element of a sample space $n(s) = 2$

Example 1:

Experiment	Outcomes	Sample Space	
Toss a coin once	Head, Tail	S = { Head, Tail }	$n = 2$
Roll a die once	1, 2, 3, 4, 5, 6	S = { 1, 2, 3, 4, 5, 6 }	$n = 6$
Toss a coin twice	HH, HT, TT, TH	S = { HH, HT, TT, TH }	$n = 4$
Birth of a baby	Boy, Girl	S = { Boy, Girl }	$n = 2$
Take a test	Pass, Fail	S = { Pass, Fail }	$n = 2$
Select a student	Male, Female	S = { Male, Female }	$n = 2$

Event

⇒ An event is a collection of one or more of the outcomes of an experiment

⇒ Consists of 2 types: simple event or compound event

Simple event (Elementary event)

⇒ An event with only one outcome for an experiment

⇒ Usually is denoted by E_i

Compound event (Composite event)

⇒ An event with more than one outcome for an experiment

Example 2:

Experiment: Toss a coin twice

Outcomes = HH, HT, TH, TT

Let E_1 be the event that both 'head' are obtained, then $E_1 = \{HH\}$

Let E_2 be the event that a 'head' followed by a 'tail', then $E_2 = \{HT\}$

Let E_3 be the event that a 'tail' followed by a 'head', then $E_3 = \{TH\}$

Let E_4 be the event that both 'tail' are obtained, then $E_4 = \{TT\}$

⇒ Simple event: E_1, E_2, E_3 and E_4

Let Q be the event that at least a 'head' is obtained, then $Q = \{HH, HT, TH\}$

Let R be the event that both toss show the same outcome, then $R = \{HH, TT\}$

⇒ Compound event: Q and R

Probability

⇒ A numerical measure of the likelihood that a specific event will occur

⇒ Is denoted by P

Notation:

$P(E_i) \equiv$ The probability that a simple event E_i will occur

✓ $P(A) \equiv$ The probability that a compound event A will occur

Two properties of probability

1. The probability of an event always lies in the range 0 to 1

⇒ $0 \leq P(E_i) \leq 1$ and $0 \leq P(A) \leq 1$

⇒ B $\equiv$ impossible event if and only if $P(B) = 0$

⇒ C $\equiv$ sure event if and only if $P(C) = 1$

not happen
✓ ← must happen

2. The sum of the probabilities of all simple events (assume n simple events) for an experiment is always 1

⇒ $\sum P(E_i) = P(E_1) + P(E_2) + \dots + P(E_n) = 1$

Example 3:

Two tosses of a coin

⇒ $P(HH) + P(HT) + P(TH) + P(TT) = 1$

Football game

⇒ $P(\text{Win}) + P(\text{Loss}) + P(\text{Tie}) = 1$

Equal likely outcomes/events

⇒ 2 or more outcomes/events that have the same probability of occurrence

Let $n(A)$ ≡ total number of outcomes belong to event A

$n(S)$ ≡ total number of outcomes for the experiment

Then

The probability of an event A , $P(A) = \frac{n(A)}{n(S)}$

Complement event

⇒ The complement of event A, denoted by $\bar{A}$ is the event that includes all the outcomes for an experiment that are not in A

⇒ $\bar{A}$ denote the event "A does not occur"

⇒ $P(\bar{A}) = 1 - P(A)$

Example 4:

A ball is chosen at random from a box containing 6 red balls, 4 white balls and 5 blue balls. Find the probability that the ball is

- a) Red b) white c) blue d) not red

Solution:

a) $P(\text{red}) = \frac{6}{15} = \frac{2}{5}$

b) $P(\text{white}) = \frac{4}{15}$

c) $P(\text{blue}) = \frac{5}{15} = \frac{1}{3}$

d) $P(\text{not red}) = 1 - P(\text{red})$
 $= 1 - \frac{2}{5} = \frac{3}{5}$

Example 5:

A die is thrown. State the sample space. Let E be the event "the number is odd" and F be the event "the number is greater than 4". Find P(E) and P(F).

Solution:

$S = \{ 1, 2, 3, 4, 5, 6 \} \rightarrow n(S) = 6$

$E = \{ 1, 3, 5 \} \rightarrow n(E) = 3$

$F = \{ 5, 6 \} \rightarrow n(F) = 2$

$P(E) = \frac{n(E)}{n(S)} = \frac{3}{6} = \frac{1}{2}$

$P(F) = \frac{n(F)}{n(S)} = \frac{2}{6} = \frac{1}{3}$

The Addition Rule

Let A and B to be two events defined in a sample space S.

The Addition Rule state that

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$

or

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Example 6:

In a group of 100 people, 40 own a cat, 25 own a dog and 15 own a cat and a dog. Find the probability that a person chosen at random will

a) Owns a cat or a dog $\rightarrow P(C \cup D)$

b) owns a cat or a dog, but not both $\rightarrow P(C \cup D) - P(C \cap D)$

C D

$$n(S) = 100$$

$$n(C) = 40 \quad P(C) = \frac{n(C)}{n(S)} = \frac{40}{100}$$

$$n(D) = 25 \quad P(D) = \frac{n(D)}{n(S)} = \frac{25}{100}$$

$$n(C \cap D) = 15 \quad P(C \cap D) = \frac{15}{100}$$

Solution:

C $\equiv$ a person owns a cat $\rightarrow P(C) = 40/100 = 0.40$

D $\equiv$ a person owns a dog $\rightarrow P(D) = 25/100 = 0.25$

C $\cap$ D $\equiv$ a person owns a cat and a dog $\rightarrow P(C \cap D) = 15/100 = 0.15$

a) $P(C \cup D) = P(C) + P(D) - P(C \cap D)$

$$= 0.40 + 0.25 - 0.15$$

$$= 0.5$$

b) P(owns a cat or a dog, but not both)

$$= P(C \cup D) - P(C \cap D)$$

$$= 0.5 - 0.15$$

$$= 0.35$$

Example 7:

$$n(S) = 20 \quad n(W) = 7 \quad n(M) = 13$$

$$P(W) = \frac{7}{20}$$

$$P(M) = \frac{13}{20}$$

In a group of 20 adults, 4 out of the 7 women and 2 out of the 13 men wear glasses. What is the probability that a person chosen at random from the group is a woman or someone who wear glasses?

Solution:

W $\equiv$ the person chosen is a woman $\rightarrow P(W) = 7/20 = 0.35$

G $\equiv$ the person wears glasses $\rightarrow P(G) = 6/20 = 0.30$

W $\cap$ G $\equiv$ woman wears glasses $\rightarrow P(W \cap G) = 4/20 = 0.20$

P(is a woman or wear glasses) $= P(W \cup G) = P(W) + P(G) - P(W \cap G)$

$$= P(W \cup G) = P(W) + P(G) - P(W \cap G)$$

$$= 0.35 + 0.30 - 0.20$$

$$= 0.45$$

$$= \frac{7}{20} + \frac{6}{20} - \frac{4}{20}$$

$$= \frac{9}{20} = 0.45$$

Example 8:

One white **die** and one black **die** are rolled. Find the probability that

- A c
- a) the white **die** shows '1' or '2' or the sum of the dice is greater than 9
- b) the white **die** shows '1' or '2' or both dice show the same number
- B
- C

Solution:

$$S \equiv \{ (1,1), (1,2), (1,3), \dots, (6,5), (6,6) \} \rightarrow n(S) = 36$$

A $\equiv$ white die shows '1' or '2'

$$\rightarrow A = \{ (1,1), (1,2), (1,3), (1,4), (1,5), (1,6), \\ (2,1), (2,2), (2,3), (2,4), (2,5), (2,6) \}$$

$$\rightarrow n(A) = 12, \quad P(A) = \frac{12}{36} = \frac{1}{3}$$

B $\equiv$ sum of dices is greater than 9, which is 10, 11, 12

$$\rightarrow B = \{ (6,6), (6,5), (6,4), (5,5), (4,6), (5,6) \}$$

$$\rightarrow P(B) = \frac{6}{36} = \frac{1}{6}$$

C $\equiv$ both dices show same number

$$\rightarrow C = \{ (1,1), (2,2), (3,3), (4,4), (5,5), (6,6) \}$$

$$\rightarrow P(C) = \frac{6}{36} = \frac{1}{6}$$

$$A \cap B = \{ \} \rightarrow P(A \cap B) = 0$$

$$a) P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$= \frac{1}{3} + \frac{1}{6} - 0$$

$$= \frac{1}{2}$$

$$A \cap C = \{ (1,1), (2,2) \} \rightarrow P(A \cap C) = 2/36 = 1/18$$

$$a) P(A \cup C) = P(A) + P(C) - P(A \cap C)$$

$$= \frac{1}{3} + \frac{1}{6} - \frac{1}{18}$$

$$= \frac{4}{9}$$

Mutually exclusive events

- ⇒ Events that **cannot occur together**
- ⇒ A and B are mutually exclusive events $\Rightarrow P(A \cap B) = 0$
- ⇒ $P(A \cup B) = P(A) + P(B)$

Example 9:

A die is thrown once. E is the event of getting even number and O is the event of getting odd number. Are the two events mutually exclusive?

Solution:

$$E = \{ 2, 4, 6 \}$$

$$O = \{ 1, 3, 5 \}$$

$$E \cap O = \{ \} \rightarrow P(E \cap O) = 0$$

Since $P(E \cap O) = 0$, therefore event E and event O are **mutually exclusive**.

Independent events

- ⇒ Two events A and B are said to be independent if the occurrence of event A does not affect the probability of the occurrence of event B
- ⇒ $P(A | B) = P(A)$ or $P(B | A) = P(B)$ or $P(A \cap B) = P(A) \times P(B)$

Dependent events

- ⇒ Two events A and B are said to be dependent if the occurrence of event A affect the probability of the occurrence of event B
- ⇒ $P(A | B) \neq P(A)$ or $P(B | A) \neq P(B)$ or $P(A \cap B) \neq P(A) \times P(B)$

Example 10:

A fair die is thrown twice. Let A be the event where a '3' is obtained on the first throw and B be the event where an odd number on the second throw. Are the two events A and B independent or dependent?

Solution:

$$A \equiv \text{'3' on 1st throw} \rightarrow P(A) = 1/6$$

$$B \equiv \text{'1', '3', '5' on 2nd throw} \rightarrow P(B) = 3/6 = 1/2$$

$$A \cap B = \{ (3,1), (3,3), (3,5) \} \rightarrow P(A \cap B) = 3/36 = 1/12$$

$$\text{Since } P(A) \times P(B) = \frac{1}{6} \times \frac{1}{2} = \frac{1}{12} = P(A \cap B)$$

therefore events A and B are independent.

Example 11:

Students take two independent tests. 30% of the students pass the test A and 60% pass the test B. Find the probability that a student selected at random will pass

- a) both tests b) only test A c) only one test

Solution:

and $\rightarrow \times$

A $\equiv$ pass test A $\rightarrow P(A) = 0.3, P(\bar{A}) = 0.7$

B $\equiv$ pass test B $\rightarrow P(B) = 0.6, P(\bar{B}) = 0.4$

a) P(pass both test)

$$\begin{aligned} &= P(A \cap B) \quad \text{and} \\ &= P(A) \times P(B) \\ &= 0.3 \times 0.6 = 0.18 \end{aligned}$$

b) P(pass only test A)

$$\begin{aligned} &= P(A \cap \bar{B}) \\ &= P(A) \times P(\bar{B}) \\ &= 0.3 \times 0.4 = 0.12 \end{aligned}$$

c) P(pass only one test) $\leftarrow$ or

$$\begin{aligned} &= P(\text{pass only test A}) + P(\text{pass only test B}) \\ &= P(A \cap \bar{B}) + P(\bar{A} \cap B) \\ &= 0.12 + P(\bar{A}) \times P(B) \\ &= 0.12 + 0.7 \times 0.6 \\ &= 0.12 + 0.42 \\ &= 0.54 \quad \times \end{aligned}$$

Note:

1. Two events A and B with $P(A) \neq 0$ and $P(B) \neq 0$ are mutually exclusive
 $\Rightarrow$ A and B are dependent events
2. Two events A and B with $P(A) \neq 0$ and $P(B) \neq 0$ are independent
 $\Rightarrow$ A and B cannot be mutually exclusive
3. Dependent events may or may not be mutually exclusive

Conditional probability

- $\Rightarrow$ probability an event A will occur given that event B has already occurred
- $\Rightarrow$ denoted by $P(A|B)$

$\Rightarrow$ If $P(B) \neq 0$ then $P(A|B) = \frac{P(A \cap B)}{P(B)}$

Example 12:

When a dice is thrown, an odd number is **obtained**. What is the probability **that** the number is a prime number?

Solution:

$O \equiv \text{odd number} \rightarrow O = \{1, 3, 5\}, P(O) = 3/6 = 1/2$

$P \equiv \text{prime number} \rightarrow P = \{2, 3, 5\}, P(P) = 3/6 = 1/2$

$ONP = \text{odd and prime number} \rightarrow ONP = \{3, 5\}, P(ONP) = 2/6 = 1/3$

$$P(P|O) = \frac{P(P \cap O)}{P(O)} = \frac{1/3}{1/2} = \frac{2}{3}$$

Example 13:

The probabilities that a student will fail accounting, mathematics, or both are $P(A) = 0.20$, $P(M) = 0.15$, and $P(A \cap M) = 0.03$ respectively. What is the probability that he will fail accounting **given** that he fail mathematics?

Solution:

$$P(A|M) = \frac{P(A \cap M)}{P(M)} = \frac{0.03}{0.15} = 0.2$$

changes only

$P(A|M)$

mutually exclusive
 $P(A \cap M) = 0.03 \neq 0$ not ME

$$P(A \cap M) = 0.03$$

$$P(A \cap M) = P(A) \times P(M)$$

$$P(A) \times P(M) = 0.2 \times 0.15 = 0.03$$

independent

The Multiplication Rule

Let A and B to be two events defined in a sample space S.

If A and B are **independent**, then $P(A \cap B) = P(A) \times P(B)$

If A and B are **dependent**, then $P(A \cap B) = P(A) \times P(B|A) = P(B) \times P(A|B)$

Example 14:

When drawing two cards from a shuffled deck, find the probability that the first card is an ace and the second card is king. Assume that the first card is **not replaced** before the second card is drawn.

Solution:

$$P(\text{ace and king}) = \frac{4}{52} \times \frac{4}{51} = 0.00603$$

Example 15:

Swee Ling and Chee Seng played ten games of chess of which six are won by Swee Ling, three are won by Chee Seng and one game ended in a tie. Now they agree to play another three games. Find the probability that

- Swee Ling and Chee Seng win alternatively,
- Chee Seng wins at least one game,
- two games will end in a tie.

$$P(\text{Swee Ling win}) = P(S) = 6/10 = 0.6$$

$$P(\text{Chee Seng win}) = P(C) = 3/10 = 0.3$$

$$P(\text{Game tie}) = P(T) = 1/10 = 0.1$$

(i) P(win alternatively)

$$= P(S \cap C \cap S) \text{ or } P(C \cap S \cap C)$$

$$= (0.6 \times 0.3 \times 0.6) + (0.3 \times 0.6 \times 0.3)$$

$$= 0.162$$

$$P(C \geq 1) = P(C=1) + P(C=2) + P(C=3)$$

(ii) P(C wins at least once)

$$= 1 - P(\text{C wins no game})$$

$$= 1 - (0.7 \times 0.7 \times 0.7)$$

$$= 0.657$$

(iii) P(two games will end in a tie)

$$= P(\bar{T} \bar{T} T) + P(T \bar{T} \bar{T}) + P(T T \bar{T})$$

$$= (0.1 \times 0.1 \times 0.9) \times 3$$

$$= 0.027$$

Probability tree

⇒ A useful way of tackling many probability problems

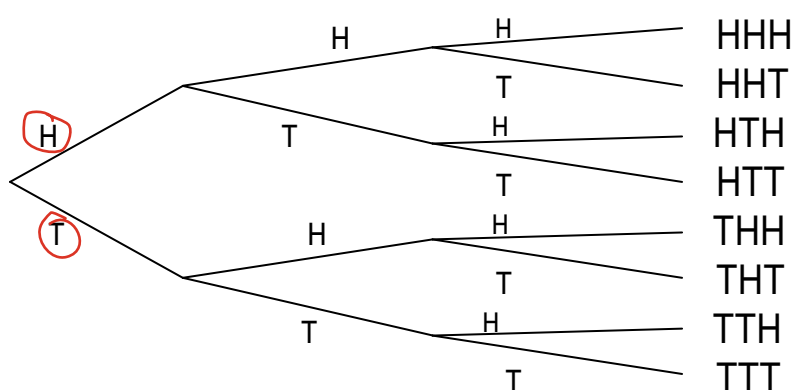
⇒ Each outcome is represented by a branch of the tree

Example 16:

Draw the probability tree for the experiment of tossing a coin 3 times. Find the sample space for this experiment.

Solution:

First toss Second toss Third toss Final outcomes

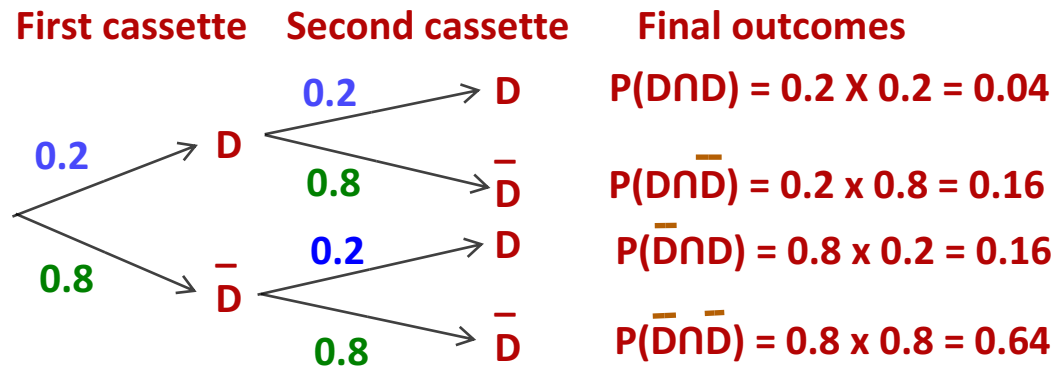


Sample space, $S = \{ HHH, HHT, HTH, HTT, THH, THT, TTH, TTT \}$

Example 17:

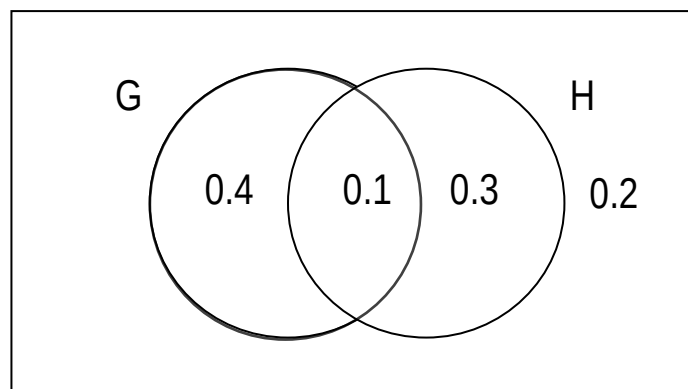
A box contains 20 cassettes, 4 of which are defective. If 2 cassettes are selected at random with replacement from this box, what is the probability that both are defective?

Solution: **$D \equiv$ defective cassette, $P(D) = 4/20 = 0.2$**



Example 18:

Given $P(G) = 0.5$, $P(H) = 0.4$ and $P(G \text{ and } H) = 0.1$. (Refer Venn diagram)



- a) Calculate
- $P(G | H)$
 - $P(H | G)$
 - $P(\bar{H})$
 - $P(G \text{ or } H)$
 - $P(G \text{ or } \bar{H})$
- b) Are the events G and H mutually exclusive? Independent?

Solution:

$$\begin{aligned}
 \text{a) i) } P(G | H) &= \frac{P(G \cap H)}{P(H)} = \frac{0.1}{0.4} = 0.25 & \text{iv) } P(G \text{ or } H) &= P(G) + P(H) - P(G \cap H) \\
 & & &= 0.5 + 0.4 - 0.1 \\
 \text{ii) } P(H | G) &= \frac{P(H \cap G)}{P(G)} = \frac{0.1}{0.5} = 0.2 & &= 0.8 \\
 \text{iii) } P(\bar{H}) &= 1 - P(H) = 1 - 0.4 = 0.6 & \text{v) } P(G \text{ or } \bar{H}) &= P(G) + P(\bar{H}) - P(G \cap \bar{H}) \\
 & & &= 0.5 + 0.6 - 0.4 \\
 & & &= 0.7
 \end{aligned}$$

b) Since $P(G \cap H) = 0.1 \neq 0$, events G and H are **not** mutually exclusive.

* Since $P(G | H) \neq P(G)$, events G and H are **not** independent.

$P(G \cap H) = 0.1$ $P(G) \times P(H) = 0.5 \times 0.4 = 0.2$ $P(G \cap H) \neq P(G) \times P(H)$ not independent.

Example 19:

A shipment of grapefruit arrived containing 10% pink seedless, 20% white seedless, 30% pink with seeds and 40% white with seeds. A grapefruit is selected at random from the shipment. Find the probability that it is

- seedless
- pink
- Pink given that it is seedless
- seedless given that it is pink

Solution:

$$S \equiv \text{seedless}, P \equiv \text{pink} \rightarrow P(P \cap S) = 0.1, P(\bar{P} \cap S) = 0.2$$

$$P(P \cap \bar{S}) = 0.3, P(\bar{P} \cap \bar{S}) = 0.4$$

$$a) P(S) = P(P \cap S) + P(\bar{P} \cap S) = 0.1 + 0.2 = 0.3$$

$$c) P(P|S) = \frac{P(P \cap S)}{P(S)} = \frac{0.1}{0.3} = \frac{1}{3}$$

$$b) P(P) = P(P \cap S) + P(P \cap \bar{S}) = 0.1 + 0.3 = 0.4$$

$$d) P(S|P) = \frac{P(P \cap S)}{P(P)} = \frac{0.1}{0.4} = \frac{1}{4}$$

Example 20:

A sample of 500 respondents was selected from a large city to determine various information concerning consumers' behaviour.

Enjoy shopping for clothing	Gender		Total
	Male (M)	Female (F)	
Yes (Y)	138	222	360
→ No (N)	106	34	140
Total	244	256	500

Contingency Table

- Suppose the respondent chosen is a female. What is the probability that she enjoys shopping for clothing?

$$P(Y|F) = \frac{P(Y \cap F)}{P(F)} = \frac{\frac{222}{500}}{\frac{256}{500}} = \frac{222}{256} = 0.8672$$

- Suppose the respondent chosen does not enjoy shopping for clothing. What is the probability that the individual is a male?

$$P(M|N) = \frac{P(M \cap N)}{P(N)} = \frac{\frac{106}{500}}{\frac{140}{500}} = \frac{106}{140} = 0.7571$$

- Construct a probability tree diagram.

	Male	Female	
Yes	138	222	360
No	106	34	140
	244	256	500

$$(i) P(\text{Yes} | \text{Female}) = \frac{P(\text{Yes} \cap \text{Female})}{P(\text{Female})} = \frac{\frac{222}{500}}{\frac{256}{500}} = \frac{222}{256} = 0.8672$$

$$(ii) P(\text{Male} | \text{No}) = \frac{P(\text{Male} \cap \text{No})}{P(\text{No})} = \frac{\frac{106}{500}}{\frac{140}{500}} = \frac{106}{140} = 0.7571$$



Exhaustive event

$\Rightarrow A_1, A_2, \dots, A_n$ are exhaustive event if $A_1 \cup A_2 \cup \dots \cup A_n = S$

Bayes theorem

Suppose $A_1, A_2, \dots, A_n$ are n mutually exclusive and exhaustive events.

Let B be an arbitrary event of S . Then, for $i = 1, 2, \dots, n$

$$P(A_i | B) = \frac{P(B | A_i)P(A_i)}{P(B)}$$

where

$$\begin{aligned} P(B) &= \sum_{i=1}^n P(B | A_i)P(A_i) \\ &= P(B | A_1)P(A_1) + P(B | A_2)P(A_2) + \dots + P(B | A_n)P(A_n) \end{aligned}$$

$\Rightarrow P(B)$ is called the total probability of event B

Example 21:

Alice, Barbara and Cathy pack biscuits in a factory. From the batch allotted to them, Alice packs 55%, Barbara 30% and Cathy 15%. The probability that Alice, Barbara and Cathy break some biscuits in a packet is 0.7, 0.2 and 0.1 respectively. What is the probability that a packet with broken biscuits was packed by Alice?

Solution:

$B \equiv$ biscuit broken

$A_1 \equiv$ packed by Alice $\rightarrow P(A_1) = 0.55 \quad \times \quad P(B | A_1) = 0.7$

$A_2 \equiv$ packed by Barbara $\rightarrow P(A_2) = 0.30 \quad \times \quad P(B | A_2) = 0.2$

$A_3 \equiv$ packed by Cathy $\rightarrow P(A_3) = 0.15 \quad \times \quad P(B | A_3) = 0.1$

$$\begin{aligned} P(B) &= P(B | A_1) P(A_1) + P(B | A_2) P(A_2) + P(B | A_3) P(A_3) \\ &= 0.7 \times 0.55 + 0.2 \times 0.30 + 0.1 \times 0.15 \\ &= 0.46 \end{aligned}$$

$$P(A_1 | B) = \frac{P(B | A_1) P(A_1)}{P(B)} = \frac{0.7(0.55)}{0.46} = 0.8370 \quad \#$$

$$P(B | A_1) = \frac{P(B \cap A_1)}{P(A_1)}$$
$$= \frac{P(A_1 \cap B)}{P(A_1)}$$
$$\# P(A_1 \cap B) = P(B | A_1) \times P(A_1)$$

Example 22:

In a certain college, 25% of the boys and 10% of the girls are studying Statistics. The girls constitute 60% of the student population. If a student is selected at random and he/she is studying Statistics, determine the probability that the student is a girl.

Solution:

G $\equiv$ the student is a girl

$$\rightarrow P(G) = 0.6, \quad P(\bar{G}) = 0.4$$

S $\equiv$ the student studying statistics $\rightarrow P(S|G) = 0.1, P(S|\bar{G}) = 0.25$

$$\begin{aligned} P(S) &= P(S|G) P(G) + P(S|\bar{G}) P(\bar{G}) \\ &= 0.1(0.6) + 0.25(0.4) = 0.16 \end{aligned}$$

$$\begin{aligned} P(S|G) &= \frac{P(S \cap G)}{P(G)} \\ P(G \cap S) &= P(S|G) \times P(G) \end{aligned}$$

$$\underline{P(G|S)} = \frac{P(S|G) P(G)}{P(S)} = \frac{0.1(0.6)}{0.16} = 0.3750$$

Arrangements

Counting Rule

If one operation consists of n steps in which the first step can be done in k_1 ways, a second steps in k_2 ways, ..., the n -th step in k_n ways, then, the number of ways performing the operation are $k_1 \times k_2 \times \cdots \times k_n$

Example 1:

If a travel agency offers special weekend trips to 7 different cities by air, rail, or bus, in how many different ways can such a trip be arranged?

Solution:

There are $7 \times 3 = 21$ ways

Example 2:

How many 3 digit numbers can be made from the integers 2, 3, 4, 5 and 6 if

- each integer is used only once?
- there is no restriction on the number of times each integer can be used?

Solution:

a) used only once

5			1st digit can filled by 5 integers.
5	4		2nd digit can filled by 4 integers.
5	4	3	3rd digit can filled by 3 integers.

There are $5 \times 4 \times 3 = 60$ ways of three digits number can be made.

b) no restriction

5	5	5	Each digit can filled by 5 integers.
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There are $5 \times 5 \times 5 = 125$ ways of three digits number can be made.

Result 1

The number of ways of arranging n unlike (different) objects in a line is $n!$

- Note:
- $n! = n(n-1)(n-2)\cdots(3)(2)(1)$
 - $0! = 1$

Example 3:

In how many ways we can arrange the letters A, B, C and D in a line?

Solution :

There are $4! = 4 \times 3 \times 2 \times 1 = 24$ ways

The arrangements are

ABCD	ABDC	ACBD	ACDB	ADCB	ADBC	BCDA	BCAD
BDAC	BDCA	BACD	BADC	CDBA	CDAB	CABD	CADB
CBAD	CBDA	DABC	DACB	DBCA	DBAC	DCAB	DCBA

Result 2

The number of ways of arranging in a line n objects of which p of first type are alike, q of a second type are alike, r of a third type are alike, and so on, is

$$\frac{n!}{p!q!r!\dots}$$

Example 4:

In how many ways can the letters of the word STATISTICS be arranged?

Solution:

There are 10 letters, with S occurs 3 times,
T occurs 3 times,
I occurs 2 times,
A occurs 1 time,
C occurs 1 time.

$$\begin{aligned}\text{Number of ways} &= \frac{10!}{3! 3! 2! 1! 1!} \\ &= 50400\end{aligned}$$

Example 5:

Five red balls, two white balls and three blue balls are arranged in a row. How many ways the balls can be arranged?

Solution:

The number of arrangements

$$= \frac{10!}{5! 2! 3!} = 2520$$

Permutation

⇒ If r objects are selected from n different objects with **attention given** to the **order of arrangement**, then, the selection is called a permutation

⇒ The number of permutations of r objects chosen from n unlike objects is

$${}^n P_r \text{ where } {}^n P_r = \frac{n!}{(n-r)!}$$

Note: ${}^n P_n = n!$

Example 6:

How many ways the **two letters** from 'A', 'B', 'C' and 'D' can be arranged?

Solution:

$$\text{There are } = {}^4 P_2 = \frac{4!}{(4-2)!} = \frac{4!}{2!} = 12 \text{ ways}$$

The arrangements are

AB BA AC CA AD DA BC CB BD DB CD DC

Example 7:

If 20 students are entered in a contest, in how many different ways can a judge award **a first prize and a second prize** to the students?

Solution:

$$\text{There are } = {}^{20} P_2 = \frac{20!}{(20-2)!} = \frac{20!}{18!} = 20(19) = 380 \text{ ways}$$

Combination

⇒ If r objects are selected from n different objects with **no attention given** to the **order of arrangement**, then, the selection is called a combination

⇒ The number of combinations of r objects chosen from n unlike objects is

$${}^n C_r \text{ where } {}^n C_r = \frac{{}^n P_r}{r!} = \frac{n!}{r!(n-r)!}$$

Example 8:

In how many ways, four letters can be chosen from the word RANDOMLY?

Solution:

$$\text{There are } = {}^8C_4 = \frac{8!}{4!(8-4)!} = \frac{8!}{4!4!} = \frac{40320}{24(24)} = 70 \text{ ways}$$

Example 9:

A committee consists of 2 ladies and 3 men are chosen from 5 ladies and 7 men. In how many ways this can be done if

- a) any lady and any man can be included?
- b) one particular man must be in the committee?
- c) two particular ladies cannot be in the committee?

Solution:

$$\begin{aligned} \text{a) 2 ladies from 5 ladies} &\rightarrow {}^5C_2 = 10 \text{ ways} \\ \text{3 men from 7 men} &\rightarrow {}^7C_3 = 35 \text{ ways} \\ \rightarrow \text{Total ways} &= 10 \times 35 = 350 \text{ ways} \end{aligned}$$

$$\begin{aligned} \text{b) 2 ladies from 5 ladies} &\rightarrow {}^5C_2 = 10 \text{ ways} \\ \text{2 men from 6 men} &\rightarrow {}^6C_2 = 15 \text{ ways} \\ \rightarrow \text{Total ways} &= 10 \times 15 = 150 \text{ ways} \end{aligned}$$

$$\begin{aligned} \text{c) 2 ladies from 3 ladies} &\rightarrow {}^3C_2 = 3 \text{ ways} \\ \text{3 men from 7 men} &\rightarrow {}^7C_3 = 35 \text{ ways} \\ \rightarrow \text{Total ways} &= 3 \times 35 = 105 \text{ ways} \end{aligned}$$

Example 10:

Find the number of ways in which 10 students can be divided into 2 groups of

- a) six and four students
- b) five students each

Solution:

$$\begin{aligned} \text{a) 6 students from 10 students in } &{}^{10}C_6 = 210 \text{ ways} \\ \text{The remaining 4 students form another group in} &1 \text{ way} \\ \rightarrow \text{Total possible selections} &= 210 \times 1 = 210 \text{ ways} \end{aligned}$$

$$\begin{aligned} \text{b) 5 students from 10 students in } &{}^{10}C_5 = 252 \text{ ways} \\ \text{The remaining 5 students form another group in} &1 \text{ way} \end{aligned}$$

But these two groups can be interchanged,
so the two groups are same from any selections.
 $\rightarrow \text{Total possible selections} = 252 / 2 = 126 \text{ ways}$

Example 11:

Three letters are selected at random from the word BIOLOGY. Find the number of selection that will contain

- a) no letter O b) only one letter O
c) both of the letters O d) any letters

Solution:

a) Number of selections without letter O

= number of ways to choose 3 letters from B, I, L, G Y

$$= {}^5C_3 = 10 \text{ ways}$$

b) Number of selections with only one letter O

= number of ways to choose 2 letters from B, I, L, G Y

$= {}^5C_2 = 10$ ways

c) Number of selections with two letters O

= number of ways to choose 1 letter from B, I, L, G Y

= ${}^5C_1 = 5$ ways

d) Number of selections with any letter

$$= 10 + 10 + 5$$

= 25 ways

Check answer:

Selection with	Possible selections
no letter O	BIL, BIG, BIY, BLG, BLY, BGY, ILG, ILY, IGY, LGY
one letter O	OBI, OBL, OBG, OBY, OIL, OIG, OIY, OLG, OLY, OGY
two letter O	OOB, OOI, OOL, OOG, OOO